THE COMPARISON DIAGRAM BETWEEN ADIC COMPLETION AND DERIVED COMPLETION FOR A PRINCIPAL IDEAL

THE STACKS PROJECT AUTHORS

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Derived completion of a complex along a principal ideal, and the complexes K_n obtained by tensoring with the map f^n. used throughout the proof, lines 26959-27000. Supplies the two objects being compared.
- Inverse limits of short exact sequences and the Mittag-Leffler condition. the author cites Homology, Lemma homology-lemma-Mittag-Leffler at proof lines 26970-26972. Justifies passing to the inverse limit and keeping exactness.
- The long exact sequence associated with a principal ideal and the f-power torsion T_f. used at proof lines 26980-27000. Supplies the third column of the diagram and the identification of the boundary terms.

Source reference. Original lemma 0H32: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21302–21302. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

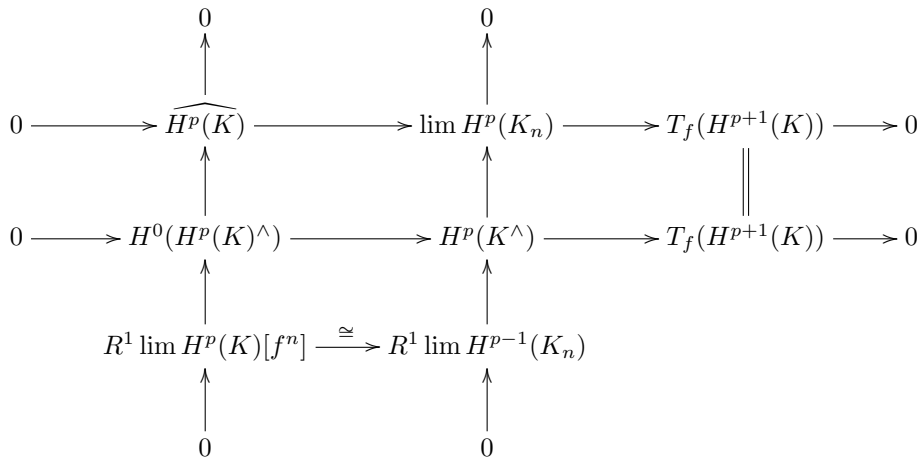
Source reference. Original lemma 0BKG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14559–14559. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0CQE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15959–15959. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02N1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3464–3464. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a ring and let $f \in A$. Let K be an object of $D(A)$. Denote $K_n = K \otimes^{\mathbf{L}}_A (A \xrightarrow{f^n} A)$. For all $p \in \mathbf{Z}$ there is a commutative diagram*



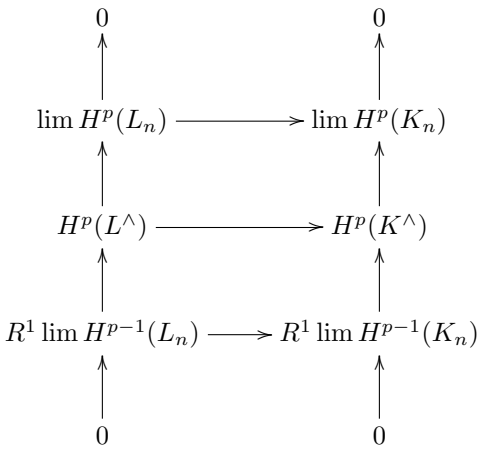
with exact rows and columns where $\widehat{H^p(K)} = \lim H^p(K)/f^n H^p(K)$ is the usual f-adic completion. The left vertical short exact sequence and the middle horizontal short exact sequence are taken from Example 0BKG The middle vertical short exact sequence is the one from Lemma 0CQE.

Proof. To construct the top horizontal short exact sequence, observe that we have the following inverse system short exact sequences

$$0 \to H^p(K)/f^n H^p(K) \to H^p(K_n) \to H^{p+1}(K)[f^n] \to 0$$

coming from the construction of K_n as a shift of the cone on $f^n : K \to K$. Taking the inverse limit of these we obtain the top horizontal short exact sequence, see Homology, Lemma 02N1.

Let us prove that we have a commutative diagram as in the lemma. We consider the map $L = \tau_{\leq p} K \to K$. Setting $L_n = L \otimes^{\mathbf{L}}_A (A \xrightarrow{f^n} A)$ we obtain a map $(L_n) \to (K_n)$ of inverse systems which induces a map of short exact sequences



Since $H^i(L) = 0$ for $i > p$ and $H^p(L) = H^p(K)$, a computation using the references in the statement of the lemma shows that $H^p(L^\wedge) = H^0(H^p(K)^\wedge)$ and that $H^p(L_n) = H^p(K)/f^n H^p(K)$. On the other hand, we have $H^{p-1}(L_n) = H^{p-1}(K_n)$ and hence we see that we get the isomorphism as indicated in the statement of the lemma since we already know the kernel of $H^0(H^p(K)^\wedge) \to \widehat{H^p(K)}$ is equal to $R^1 \lim H^p(K)[f^n]$. We omit the verification that the rightmost square in the diagram commutes if we define the top row by the construction in the first paragraph of the proof. $\square$